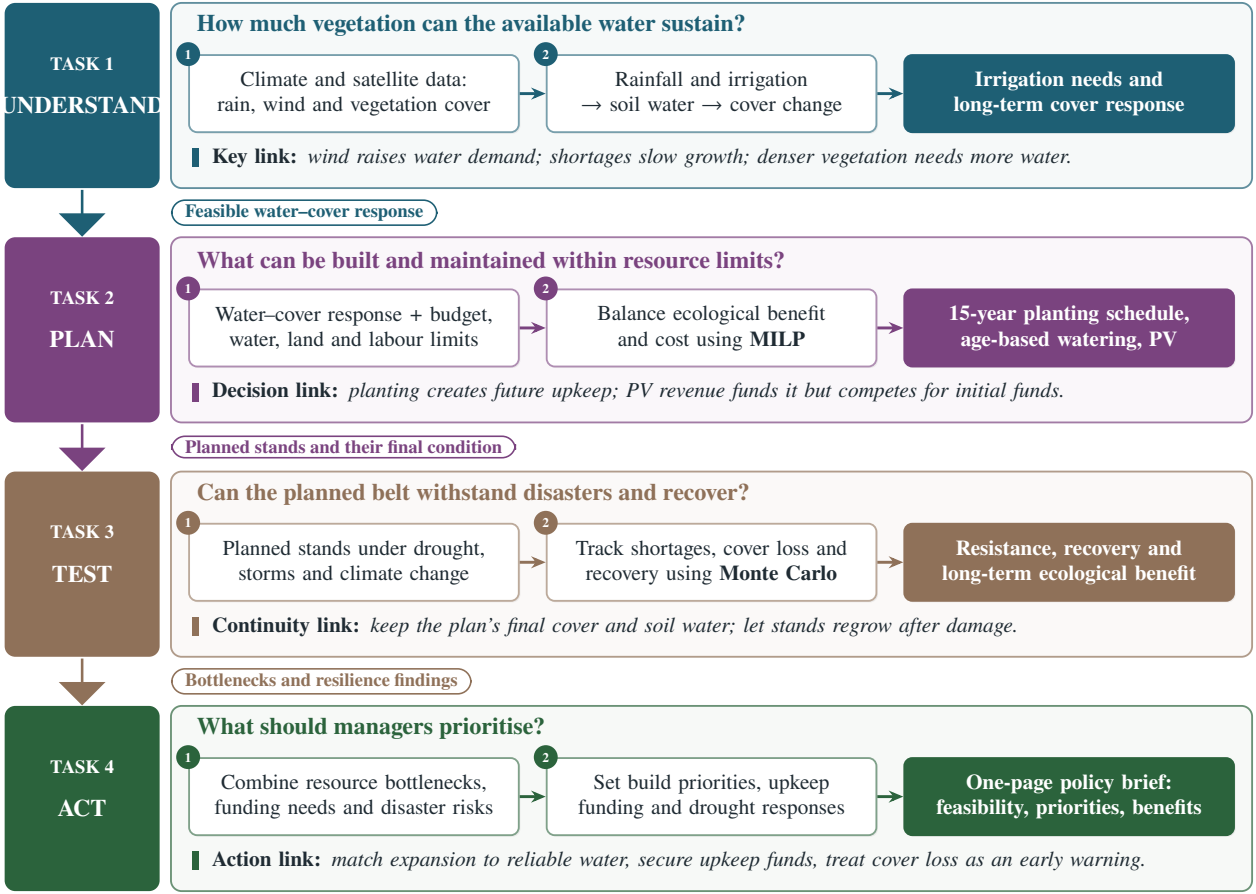

TASK 1
UNDERSTAND
How much vegetation can the available water sustain?
1
Climate and satellite data: rain, wind and vegetation cover
2
Rainfall and irrigation → soil water → cover change
Irrigation needs and long-term cover response
Key link: wind raises water demand; shortages slow growth; denser vegetation needs more water.
Feasible water–cover response
TASK 2
PLAN
What can be built and maintained within resource limits?
1
Water–cover response + budget, water, land and labour limits
2
Balance ecological benefit and cost using MILP
15-year planting schedule, age-based watering, PV
Decision link: planting creates future upkeep; PV revenue funds it but competes for initial funds.
Planned stands and their final condition
TASK 3
TEST
Can the planned belt withstand disasters and recover?
1
Planned stands under drought, storms and climate change
2
Track shortages, cover loss and recovery using Monte Carlo
Resistance, recovery and long-term ecological benefit
Continuity link: keep the plan's final cover and soil water; let stands regrow after damage.
Bottlenecks and resilience findings
TASK 4
ACT
What should managers prioritise?
1
Combine resource bottlenecks, funding needs and disaster risks
2
Set build priorities, upkeep funding and drought responses
One-page policy brief: feasibility, priorities, benefits
Action link: match expansion to reliable water, secure upkeep funds, treat cover loss as an early warning.